

Date: 4.23.2025

Name: Mari Zampa

Lab Partners: Kaimana Kau, Jack Merrill, Ryan Finley

Lab 3: Rheometry

Overview

This experiment shows how fluids that don't follow Newton's law behave under different forces. We used a rheometer to quantify viscosity, yield stress, and their viscoelastic response. Ketchup was our main test fluid for rotational (shear-rate and shear-stress controlled) and oscillatory (amplitude sweep) measurements. Body wash, toothpaste, honey, and a cornstarch (water mix) were analysed to see whether they thin out or thicken when sheared, and to note any yield behavior for comparison.

Setup

Figure 1 shows the experimental setup.

- Fluid 1: ketchup
- Fluids 2 - 4: Body wash, toothpaste, and honey
- Fluid 5: Cornstarch–water suspension (prepared at ~2:1 starch-to-water ratio to exhibit shear-thickening)
- MCR 92 Rheometer
- Stir sticks and cups



Figure 1: In this experiment, the following tests were run on ketchup:

1. Shear rate controlled rotational test: Applied shear rates from 0.1 to 60 s^{-1} to map the flow curve and calculate apparent viscosity.
2. Shear stress controlled rotational test: Increased shear stress up to 350 Pa to identify the yield stress and post-yield flow.
3. Amplitude sweep oscillatory test: Imposed sinusoidal strains from 10^{-4} to 1 rad at a fixed frequency of 10 Hz to locate the linear viscoelastic region (LVER).

Fluids were administered at varying rates during the tests. Observations on shear thinning or thickening behavior, the presence of a yield point, and relative viscosities were noted.

Raw data

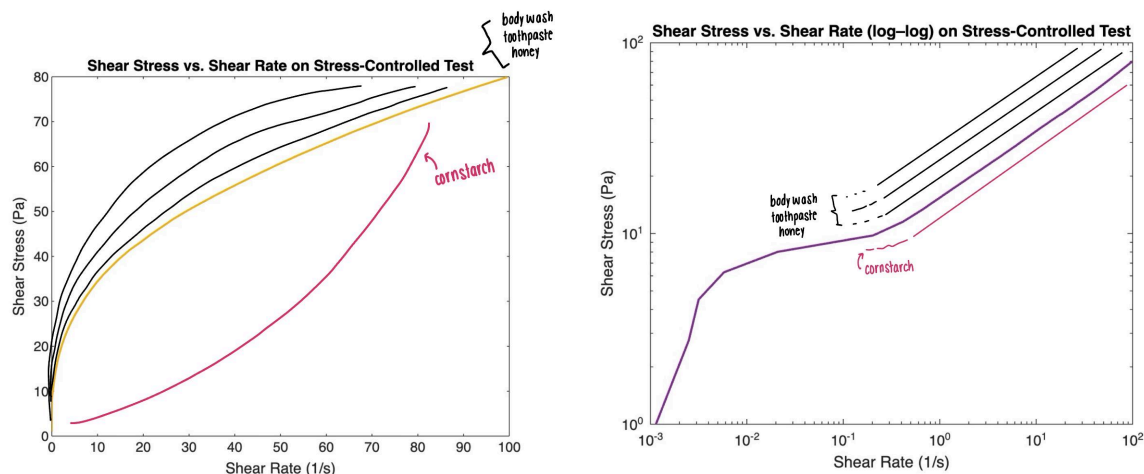


Figure 2 shows how shear stress changes with shear rate for all five samples, drawn on a regular scale (left) and on log-log axes (right). The ketchup data is plotted in yellow from our controlled-rate test. Based on our hand-tests, we've positioned body wash, toothpaste, and honey alongside the ketchup curve. These three all become less viscous as you increase the shear rate (shear-thinning). By contrast, the cornstarch gets thicker when sheared, which is why its curve bends upward sharply.

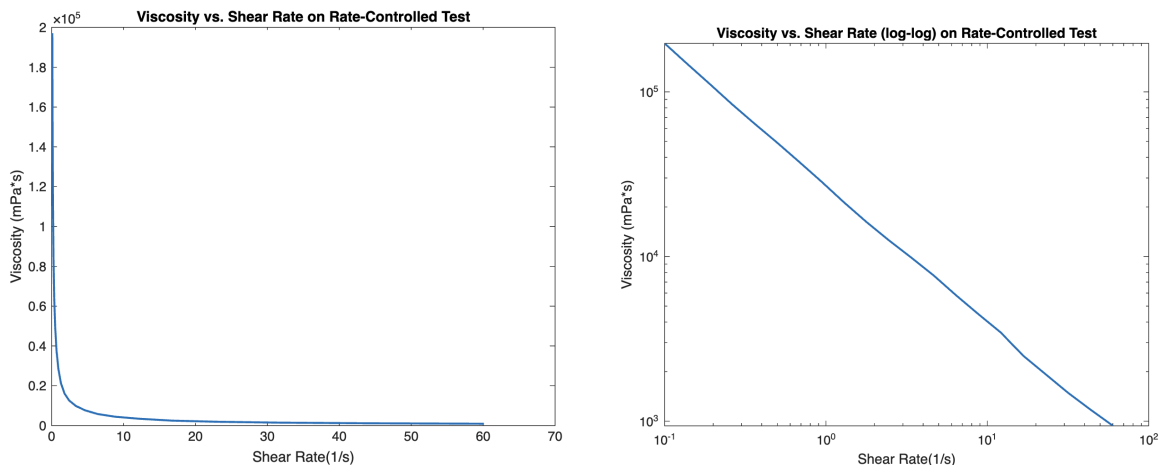


Figure 3 shows ketchup's viscosity plotted against shear rate on a linear scale (left) and a log-log scale (right). As the shear rate increases, the viscosity falls off sharply, confirming that ketchup behaves as a shear-thinning fluid.

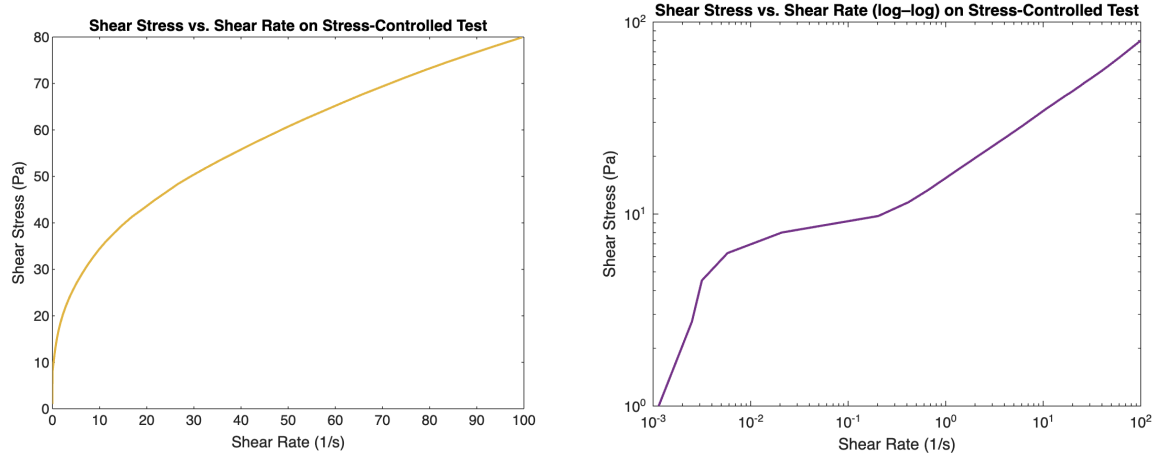


Figure 4 shows ketchup's shear stress vs shear rate in the stress-controlled rotational test, with a linear-scale graph on the left and a log-log graph on the right. The linear plots resemble those from the rate-controlled test, but the log-log curve here is slightly curved rather than perfectly straight. This happens because the stress-controlled method increases force in small steps, giving the ketchup time to adjust and showing minor yield effects that are less obvious under fixed-rate control.

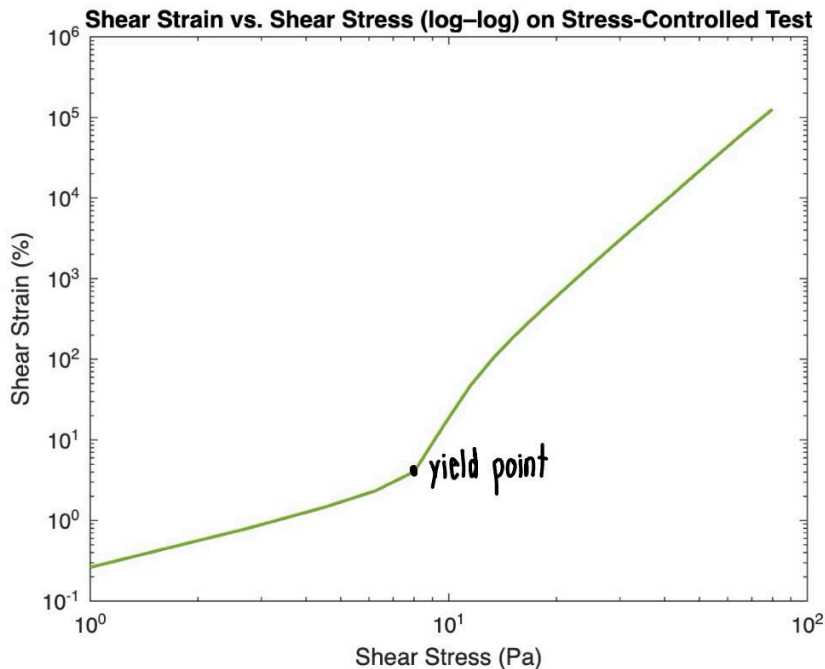


Figure 5 plots shear stress vs shear strain for ketchup in the stress-controlled test on log-log axes. By fitting straight lines to the initial (elastic) and later (flow) regions and finding where they intersect, we locate the yield point at about 3 % strain and 6.5 Pa stress

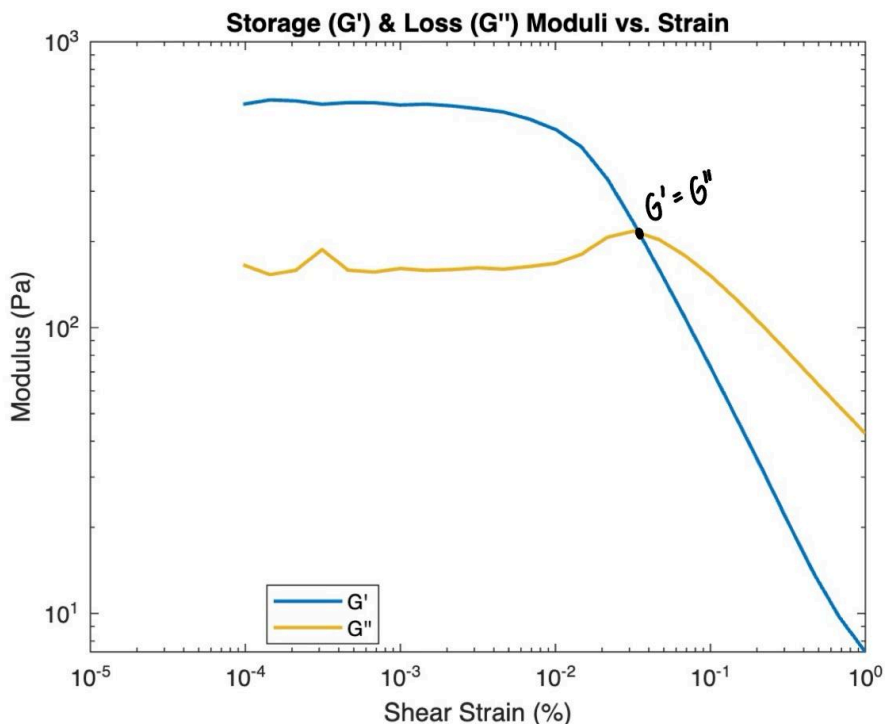


Figure 6 plots the storage modulus (G') and loss modulus (G'') vs shear strain during an oscillatory amplitude sweep. At small strains, G' is much higher than G'' , indicating the material behaves like a solid. As strain increases, the two curves meet and then G'' rises above G' . That crossover marks the end of the viscoelastic region: below it, the fluid acts solid-like, and above it, it acts liquid-like

Discussion

In this experiment, we looked at how several everyday fluids flow under different stresses, focusing mainly on ketchup. We found that ketchup is shear-thinning. Its viscosity drops as the shear rate increases, and it only begins to yield at around 6.5 Pa when strained about 3 %. In oscillatory measurements, ketchup behaved like a linear viscoelastic material up to around 0.02 strain. At about 217 Pa of strain, the elastic and viscous responses crossed over, showing its shift from solid to liquid behavior. When we compared the other fluids, body wash, toothpaste, and honey all showed the same shear-thinning trend, but the cornstarch-water mixture thickened as shear increased, showing classic shear-thickening behavior.